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$u, v \in \mathbb{R}^d$: $\|u - v\|_2 = \sqrt{\sum (x_i - v_i)^2}$

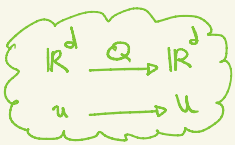
- $\|u\|_Q$ with :

  Q - positive definite, squared and symmetrical ($Q_{d \times d}$)

  * the $x^T Q x \geq 0$: $x^T Q x = 0 \iff x = 0$

  * All λ of Q are real and positive eigenvalues

→ $Q = R \Lambda R^T = \underbrace{(R\sqrt{\Lambda})}_{T}\underbrace{(\sqrt{\Lambda} R^T)}_{T^T}$



- Rotation : $R \cdot R = I$

- $\sqrt{\Lambda} = \begin{pmatrix} \sqrt{\lambda_1} & & \\ & \ddots & \\ & & \sqrt{\lambda_n} \end{pmatrix}$

$u = T \cdot u$

$\|u\| = \|u\|_Q$

→ Quadratic forms are differentiable.

* $f(u) = x^T Q x = \sum_{i=1}^{d}\sum_{j=1}^{d} Q_{ij} \cdot x_i \cdot x_j = Q_{11} x_1^2 + Q_{12} x_1 x_2 + \dots + Q_{1d} x_1 x_d +$

$\vdots$

$Q_{i1} x_i x_1 + Q_{i2} x_i x_2 + \dots + Q_{id} x_i x_d +$

$\vdots$

$Q_{d1} x_d \cdot x_1 + Q_{d2} x_d \cdot x_2 + \dots + Q_{dd} x_d^2$

x_i

* $\nabla f(x) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \vdots \\ \frac{\partial f(x)}{\partial x_d} \end{bmatrix}_d$

* $\frac{\partial f(x)}{\partial x_i} = Q_{i1} x_1 + Q_{i2} x_2 + \dots + Q_{ii} x_i + \dots Q_{id} x_d +$

$Q_{1i} x_1 + Q_{2i} x_2 + \dots + Q_{ii} x_i + Q_{di} x_d$

$\underbrace{Q(i,:) \cdot x}_{\text{Row}} + \underbrace{Q(:,i) \cdot x}_{\text{Column}} = (Q(i,:) + Q^T(:,i)) \cdot x$

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$$\nabla f(x) = (Q + Q^T).x$$

if $Q = Q^T \Rightarrow \nabla f(x) = 2Qx$

Exemple :

$$Q = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 3 & 1 \\ 1 & 2 & 3 \end{bmatrix} \qquad f(x) = x^T Q x = x^T \left(x_1 Q(:,1) + x_2 Q(:,2) + x_3 Q(:,3) \right)$$

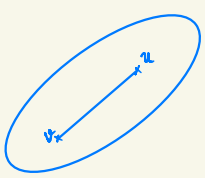
$$= Q_{11} x_1 x_1 + Q_{12} x_1 x_2 + Q_{13} x_1 x_3 + Q_{21} x_2 x_1 + Q_{22} x_2 x_2 + Q_{23} x_2 x_3 + Q_{31} x_3 x_1 + Q_{32} x_3 x_2 + Q_{33} x_3 x_3$$

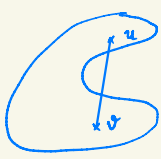
$$\frac{\partial f(x)}{\partial x_1} = Q_{11} x_1 + Q_{12} x_2 + Q_{13} x_3 + Q_{11} x_1 + Q_{21} x_2 + Q_{31} x_3 = Q(1,:).x + Q(:,1).x = \left(Q(1,:) + Q^T(:,1)\right).x$$

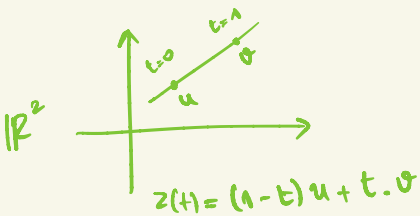
$$\frac{\partial f(x)}{\partial x_2} = \left(Q(2,:) + Q^T(:,2)\right).x$$

$$\frac{\partial f(x)}{\partial x_3} = \left(Q(3,:) + Q^T(:,3)\right).x$$

- **Convexity :**







C is convex if: $\forall u, v \in C$, $\forall t \in [0,1]$:

$$z(t) = (1-t)u + t.v \in C$$

Convex combination.

$z(t) = (1-t)u + t.v$, $t \in [0,1]$

$\begin{cases} z(0) = u \\ z(1) = v \end{cases}$

- $H(a,b) = \{u \mid a.u \leq b\}$ (Half space)

$a, x \in \mathbb{R}^d$

$b \in \mathbb{R}$

$a = [1,2]$

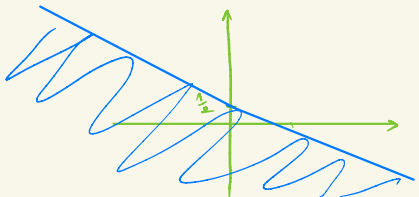
$b = 1$

$\mathcal{L}(a,b) = \{u \mid a.u = b\}$

$1.u_1 + 2.u_2 = 1$

$u_1 = 0 \rightarrow u_2 = \frac{1}{2}$

$u_1 = 1 \rightarrow u_2 = 0$



Made with Goodnotes

Proof:

1- $u, v \in H(a,b)$:

$t \in [0,1]$: $(1-t) \cdot [a \cdot u \leq b]$

$t \quad [a \cdot v \leq b]$

$$\underbrace{(1-t)a \cdot u + t a \cdot v}_{a\underbrace{((1-t)u + tv)}_{z(t)}} \leq \underbrace{(1-t)b + tb}_{b}$$

2- $H(0,r) = \{ x \in \mathbb{R}^d \mid \|u\| \leq r \}$ – is convex

$u, v \in H(0,r)$: $\|u\| \leq r$

$\|v\| \leq r$

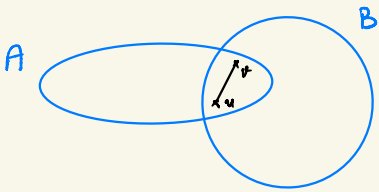
$t \in [0,1]$: $z(t) = (1-t)u + t \cdot v$

$\|z(t)\| = \|(1-t)u + tv\| \leq |1-t| \cdot \|u\| + |t| \cdot \|v\| \leq (1-t)r + tr$

$\iff \|z(t)\| \leq r$

– A, B convex $\implies A \cap B$ is convex

$\implies A \cup B = ?$ Not convex (in general)



$\mu, v \in A \cap B \iff u, v \in A$ and $u, v \in B$

$\implies [u,v] \subseteq A$ and $[u,v] \subseteq B \implies [u,v] \subseteq A \cap B$

f – Linear (affine) Transform

$f : \mathbb{R}^d \longrightarrow \mathbb{R}^m$

$u \longrightarrow \underset{m \times d}{A} u + \underset{m \times 1}{b}$

$D = f(C)$

image

↓

is Convex!?

Proof:

$C \subseteq \mathbb{R}^d$, $\exists u, v \in C$, $U, V \in D$, $t \in [0,1]$

convex

$f(u) = U$

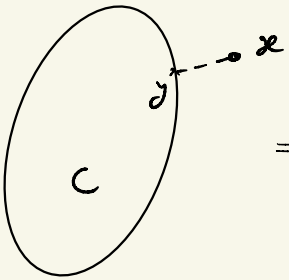
$f(v) = V$

$z = (1-t)U + tV = (1-t)(Au+b) + t(Av+b)$

$= A((1-t)u + tv) + (1-t+t)b$

$z = Az + b \in D$!

Made with Goodnotes



$\Longrightarrow$ Simple



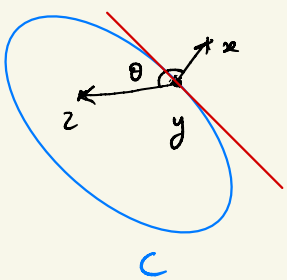
$\Longrightarrow$ Complicated

$$y \in C,\ \|x-y\| = \inf_{u \in C} \|u-x\|,\ y = \text{proj}(x,C)$$

Theorem : C be a closed convex set in $\mathbb{R}^d$ then :

1) $y = \text{proj}(x,C) \iff \forall z \in C\ (x-y)\cdot(z-y) \leq 0$

2) y is unique.



$$(x-y)\cdot(z-y) = \|x-y\| \times \|z-y\| \cdot \cos(\theta)$$

Proof : Let $y \in C$ s.t $\|x-y\|$ the least

Let $x \in C$, $0 \leq t \leq 1$

$$z(t) = (1-t)y + tz$$

$$\|x-y\|^2 \leq \|x-z(t)\|^2 = \|(x-y)+t(y-z)\|^2 =$$

??? $\|x-y\|^2 + t^2\|z-y\|^2 + 2t(x-y)(y-z)$

$$0 \leq \underbrace{\tfrac{t}{2}\|y-z\|^2 + (x-y)(y-z)}_{???} \xrightarrow[t \to 0]{}$$

y s.t $\forall z \in C$

$(x-y)(z-y) \leq 0$

$\|x-z\|^2 = \|(x-y)-(z-y)\|^2$???

$\|x-z\|^2 - 2(x-y)(z-y) + \|x-y\|^2 \geq \|x-y\|^2$

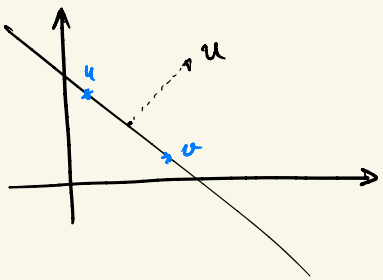
???

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Exercice 8

$H(a,b) = \{ u \in \mathbb{R}^d \mid a \cdot u \leq b \}$

$u \notin H(a,b)$ Find $y = \text{proj}(u, H)$



$$\begin{cases} a \cdot u = b \\ a \cdot v = b \end{cases} \Rightarrow \{ a \cdot (u - v) = 0$$

$y(t) = u + at \Rightarrow \exists t_0$ s.t. $a \cdot y(t_0) = b$

$t \in \mathbb{R}$

$$a \cdot (u + a t_0) = b$$

$$\boxed{t_0 = \frac{b - a \cdot u}{a \cdot a}} < 0 \quad a \cdot u > b$$

$z \in H$: $(z - y(t_0))(u - y(t_0)) = (z - y(t_0)) \cdot (-t_0 a)$

$= -t_0 (a \cdot z - a \cdot y(t_0)) = -t_0 (a \cdot z - b) \leq 0$

($a \cdot y(t_0) = b$, $-t_0 \geq 0$, $a \cdot z - b \leq 0$)

$$\text{dist}(u, H) = \| u - y(t_0) \| = \| a \left(\frac{b - au}{\|a\|^2} \right) \| = \frac{|b - a \cdot u|}{\|a\|}$$

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